Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - II

Course Code: BCA104-2 **Max Marks: 100**

Course Name: DISCRETE MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Demonstrate a working knowledge of set notation and elementary set theory and recognize the connection between set operations and logic.

CO2: Apply the different properties of injections, surjections, bijections, compositions, and inverse functions.

CO3: Understand different types of matrices and operations among matrices.

CO4: Determine when a relation is reflexive, symmetric, antisymmetric, transitive, apply the properties of equivalence relations and partial orderings, and explain the connection between equivalence relations.

CO5: Effective express the elementary concepts and results of Number theory.

Section A **3*8=24**

1)	Examine the algorithm for finding the least common multiple (LCM) of two integers using their greatest common divisor (GCD). Using this, find the LCM of 140 and 350.	**3**	(CO1, L4)
2) a)	Describe how to prove that the set of rational numbers between 0 and 1 is countable.	**3**	(CO2, L2)
	OR		
b)	Discuss the properties of a function that is both an injection and a surjection, and illustrate your answer with an example.	**3**	(CO2, L2)
3)	Explain matrix multiplication and provide an example using two 2x2 matrices.	**3**	(CO3, L2)
4)	Examine and differentiate between a symmetric relation and an antisymmetric relation using examples from the given set A = {1, 2, 3} where R = {(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)}.	**3**	(CO4, L4)
5) a)	Discuss how you would determine if a given function f: A → B is bijective by examining its properties and provide an example to illustrate your explanation.	**3**	(CO5, L2)
	OR		
b)	Explain the Euclidean algorithm for finding the greatest common divisor of two integers and use it to find ν, where ν is the gcd(576, 459).	**3**	(CO5, L2)
6)	Determine whether the set of all rational numbers between 0 and 1 (inclusive of 0 but exclusive of 1) is countable. Justify your answer.	**3**	(CO1, L3)

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7) a)	Examine and categorize the given function f: A → B defined by f(x) = 2x + 1 for x A into one of the following types: injection, surjection, bijection, or neither. Justify your answer.	**3**	(CO2, L4)
	OR		
b)	Examine and differentiate between an injective function and a surjective function using two distinct examples for each. Provide justification for your classification.	**3**	(CO2, L4)
8)	Examine and compare the properties of a matrix and its transpose, specifically focusing on symmetry and their roles in defining symmetric relations. Provide an example of a 3x3 matrix that represents a symmetric relation and explain your reasoning.	**3**	(CO3, L4)
	Section B		**8*7=56**
9)	Examine and differentiate between reflexive, symmetric, antisymmetric, and transitive properties in relations, using specific examples to illustrate your analysis. Additionally, prove or disprove whether the following relation R on the set A = {1, 2, 3, 4} is an equivalence relation: R = {(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (3, 4)}. If it is not an equivalence relation, identify which properties are violated and provide a counterexample for each. Finally, discuss the implications of these properties in defining partial orderings.	**8**	(CO4, L4)
10) a)	Determine and prove that two integers a and b are coprime using the Euclidean algorithm. Then, apply your method to find all pairs of positive integers less than 10 which are coprime.	**8**	(CO5, L3)
	OR		
b)	Consider two relations, and , defined over the set = {1, 2, 3, 4} where = {(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (2, 3), (3, 2)} and = {(1, 1), (2, 2), (3, 3), (4, 4), (3, 4), (4, 3)}. Determine the following: (i) The symmetric difference of relations and . (ii) The relation obtained by composing and . (iii) Whether the composed relation from part (ii) is an equivalence relation. If it is not, identify which properties (if any) are missing. (iv) Construct the matrix representation of each of the relations and .	**8**	(CO5, L3)
11)	Demonstrate how to use set theory and logic to solve the following problem: Given two sets A = {x ℤ | x mod 4 = 0} and B = {y ℤ | y mod 6 = 0}, find the intersection of sets A and B, denoted as A ∩ B. Further, determine which integers between 1 and 30 (inclusive) belong to both A and B. Construct a Venn diagram to illustrate your solution.	**8**	(CO1, L3)
12) a)	Determine whether the following sets are countable and provide rigorous proofs for your answers. Your proof should demonstrate an understanding of bijections, injections, or surjections as necessary. (i) The set of all positive rational numbers that can be expressed with denominators less than 10. (ii) The set of all polynomials with integer coefficients and degree at most 3. (iii) The set of all infinite sequences of 0s and 1s.	**8**	(CO2, L3)
	OR		
b)	Using properties of congruences and basic number theory, solve the following problem: Determine whether the function : ℤ → ℤ defined by () = 5 + 3 is an injection, a surjection, or both. Justify your answer using appropriate examples and mathematical arguments.	**8**	(CO2, L3)

13)	Discuss the properties and types of matrices that are commonly used in discrete mathematics. Further, illustrate these concepts by explaining how matrix operations such as addition, multiplication, and finding the inverse can be applied to solve a system of linear congruences using a specific example. Use the modulus operation in your explanation.	**8**	(CO3, L2)
14) a)	Given the sets = {1, 2, 3, 4} and = {a, b, c}, let relations 1 and 2 be defined on these sets as follows: 1 = {(1, a), (2, b), (3, c), (4, a)} 2 = {(1, b), (2, c), (3, a), (4, b)}. Determine the following properties for each relation and their combination: (i) Reflexive (ii) Symmetric (iii) Antisymmetric (iv) Transitive. Then, consider 1 2 and find its properties from the options provided. Show your work for each step.	**8**	(CO4, L3)
	OR		
b)	Determine and prove if the given relations are reflexive, symmetric, antisymmetric, or transitive over their respective sets. If they possess any of these properties, justify your answer with examples from the set. Additionally, if applicable, construct an example of a relation that satisfies only one of these properties among the given options. (i) Let be defined on = {1, 2, 3} by = {(1,1), (2,2), (3,3)}. (ii) Let be defined on = {a, b, c, d} by = {(a,a), (b,b), (c,c), (d,d), (a,c), (b,d), (c,d)}. (iii) Construct a relation on the set = {1, 2, 3, 4} that is only reflexive. Provide examples to support your assertions.	**8**	(CO4, L3)
15)	Explain with examples how to determine if a given binary relation on a set is symmetric and transitive. Also, discuss the difference between these two properties and classify the following relations as either symmetric or transitive or both: (i) On the set {1, 2, 3}, let R = {(1, 2), (2, 1), (3, 3)}. (ii) On the same set, define S = {(1, 1), (2, 2), (3, 3), (1, 3), (3, 1)}.	**8**	(CO5, L2)
	Section C		**10*2=20**
16)	Consider the sets A = {1, 2, 3} and B = {1, 2, 3, 4}. Let R_1 and R_2 be relations on these sets defined as follows: R_1 = {(1,1), (2,2), (3,3)} and R_2 = {(1,1), (1,2), (1,3), (1,4)}. Analyse the following operations on these relations and provide their resulting relations in roster form. Additionally, determine if each result is reflexive or symmetric: (i) R_1 R_2 (ii) $R_1 \cap R_2$ (iii) R_1 R_2 (iv) R_1 - R_2	**10**	(CO1, L4)
17) a)	Examine and analyze how to determine if a function f: Z → Z defined by f(x) = x^3 + ax + b is injective, surjective, or bijective for integers a and b. Provide examples of specific values for a and b to illustrate your analysis.	**10**	(CO2, L4)
	OR		
b)	Consider the function : → defined by the rule () = 2 + 1 for all , where and are subsets of real numbers. Determine if is an injection, a surjection, or a bijection from to by analyzing its properties. Additionally, examine the composition of functions where : → and () = 3 − 2 for all . Finally, test if there exists an inverse function −1 and find it if possible.	**10**	(CO2, L4)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate

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L6 - Create